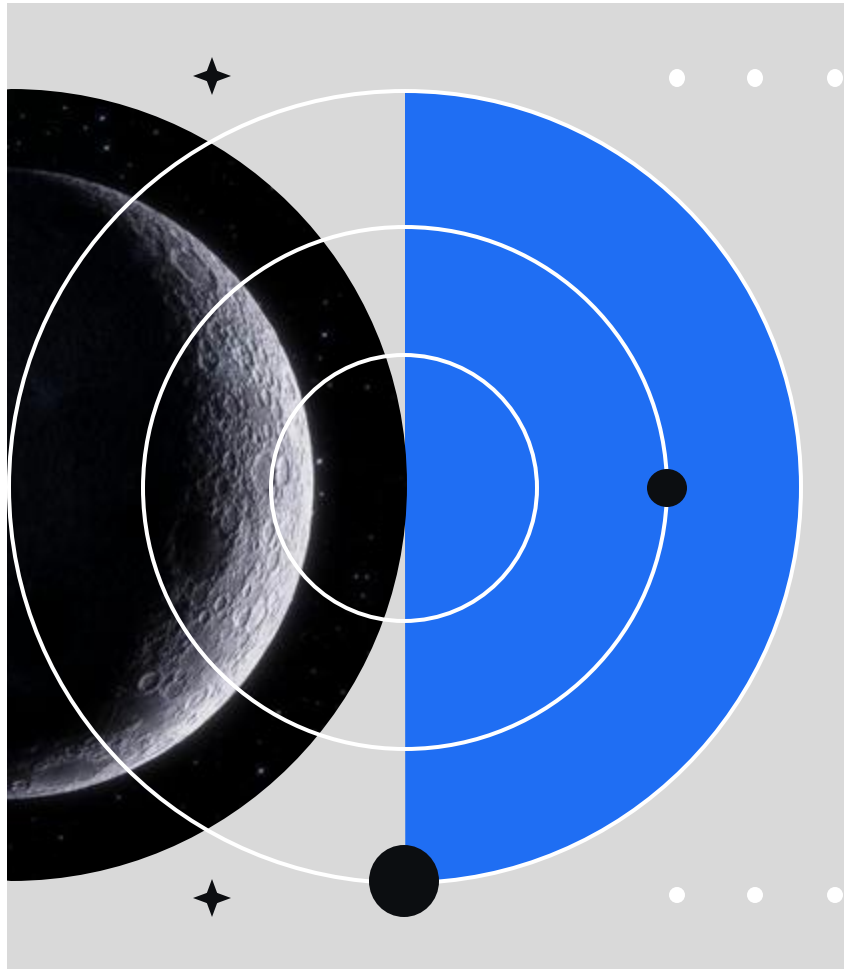




SOLAR SAIL FEASIBILITY FOR LUNAR MISSIONS

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INTRODUCTION

- Solar sails are recent technology first tested in 2010 (IKAROS). They are thin materials that spreads over a large area to capture maximum solar radiation from the sun and produce a small amount of thrust when solar radiation. They are considered as promising propulsion technology to expand the frontiers of the space missions.
- This research work aims to investigate the feasibility of solar sail interplanetary fly-by missions from a parking orbit around earth by calculating the required area & volume of a solar sail and determining the possibility of fitting in a 2U form factor.

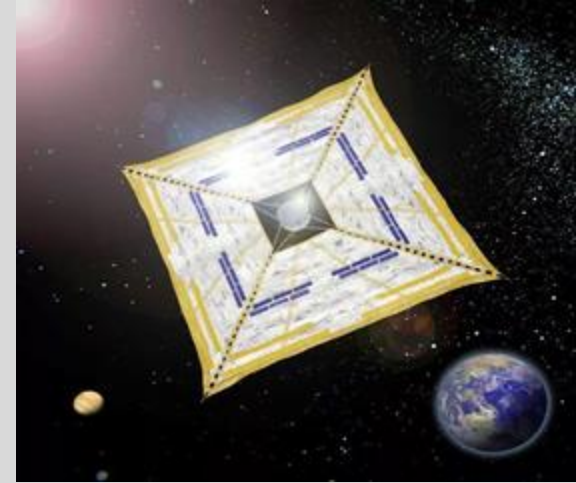
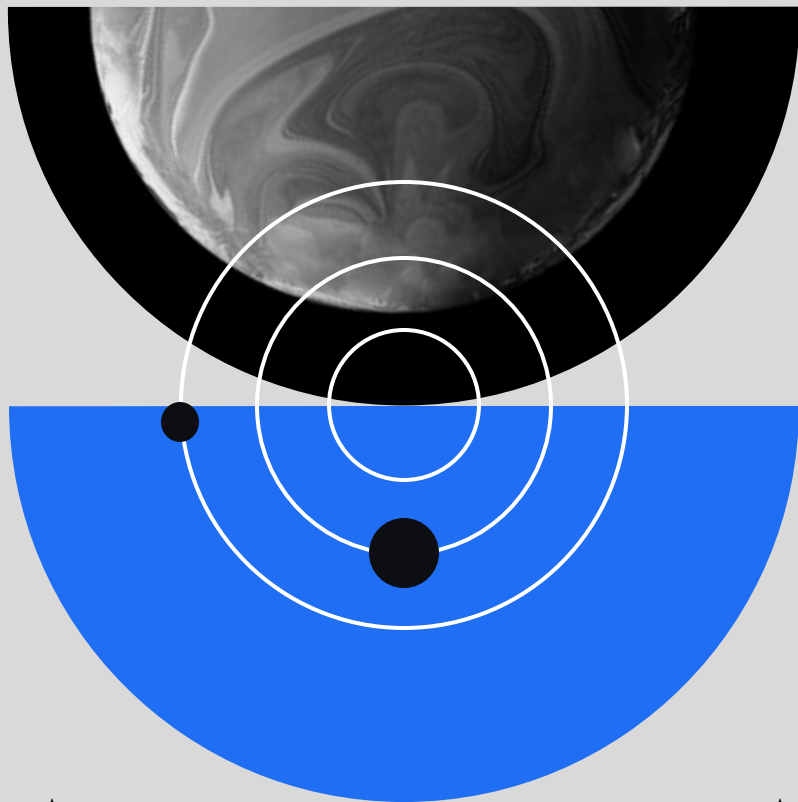


Figure 1 IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun) spacecraft designed to use solar sailing principle for interplanetary missions .



01

Problem statement

Is a 2U CubeSat Form factor with current solar sail propulsion technology feasible to execute an interplanetary mission?

If not, what is the smallest feasible form factor? And what is its area?



MOTIVATION



Practical

- Current technology considered
- Capable of immediate impact



Applications

- Help grow the space market
- Aid missions (Artemis)



Cost

- Low costs due to size
- Multiple tests to collect data at low cost

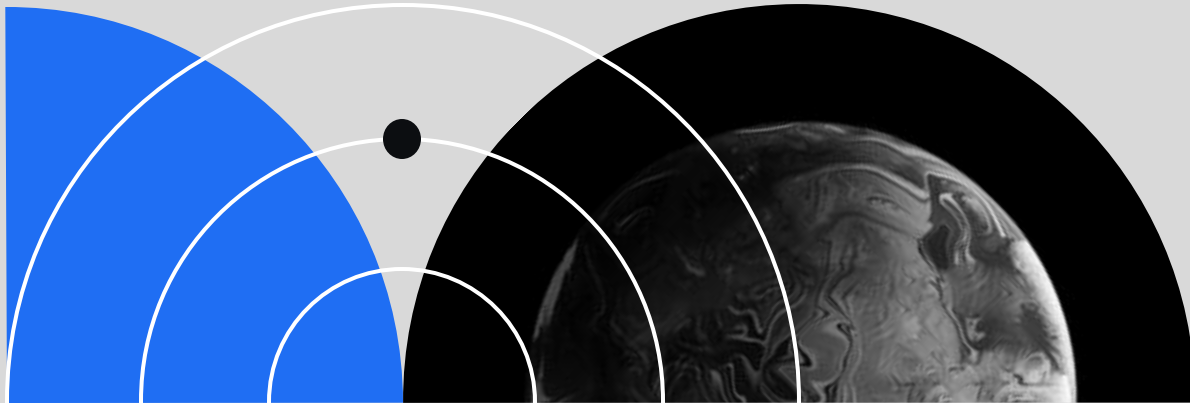


Space for all

- Allows researchers all over the world to explore further
- Space more accessible

02

METHODOLOGY



MODELLING & SIMULATION

- Initial condition: circular parking orbit at 800 km
- Initial true anomaly earth: 49°
- Assumptions:
 - Everything rotates on single plane
 - Restricted 3 body problem
 - Moon's orbit around Earth is perfectly circular
 - Earth's orbit around Earth is perfectly circular
 - Drag neglected
- Maximum trajectory time is limited to 1 year
- Trajectory is modelled and simulated in MATLAB.
Simulation consists of orbit propagation of the restricted three-body problem with solar sails applying force over multiple orbits of earth to increase orbit from 800 km to the target orbit.

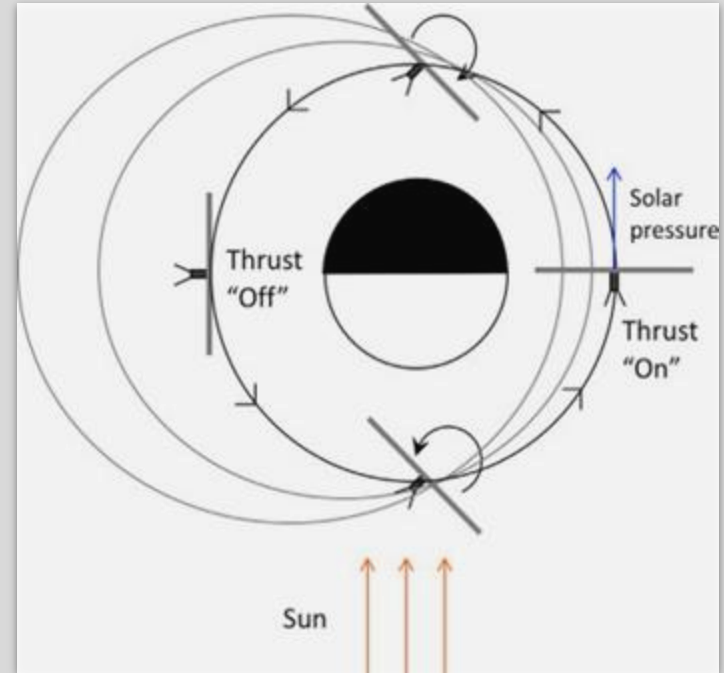


Figure 2: Illustration of planned solar thrust.

EQUATIONS OF MOTION

Equations of Motion (EOMs) for the Restricted 3-BP
with additional force from solar radiation absorption and reflection

$$\frac{d^2x}{dt^2} = -Gm_{\oplus} \frac{x(t) - x_{\oplus}}{r_{\oplus}^3} - Gm_{\zeta} \frac{x(t) - x_{\zeta}}{r_{\zeta}^3} + \frac{F_{sr,x}}{m_{sc}}$$

$$\frac{d^2y}{dt^2} = -Gm_{\oplus} \frac{y(t) - y_{\oplus}}{r_{\oplus}^3} - Gm_{\zeta} \frac{y(t) - y_{\zeta}}{r_{\zeta}^3} + \frac{F_{sr,y}}{m_{sc}}$$

$$r_{\oplus} = \sqrt{(x(t) - x_{\oplus})^2 + (y(t) - y_{\oplus})^2}$$

$$r_{\zeta} = \sqrt{(x(t) - x_{\zeta})^2 + (y(t) - y_{\zeta})^2}$$



FORCING FUNCTION

Force due to solar radiation pressure normal to the solar sail is calculated following Equation 8-43 in Vallado [4]:

$$\vec{F}_{srp} = - p_{srp} C_R A_{\odot} \frac{\vec{r}_{\oplus\odot}}{|\vec{r}_{\oplus\odot}|}$$

where

- p_{srp} is the constant solar radiation pressure $4.57 \times 10^{-6} \text{ N/m}^2$ ("Vallado")
- C_R is the reflectivity of the sail (assumed constant 1.7)
- $A_{\odot}$ is the area exposed to solar radiation
- $\vec{r}_{\oplus\odot}/|\vec{r}_{\oplus\odot}|$ is the unit vector pointing in the direction of the sun's rays.
- "Ode89" solver is used on matlab.

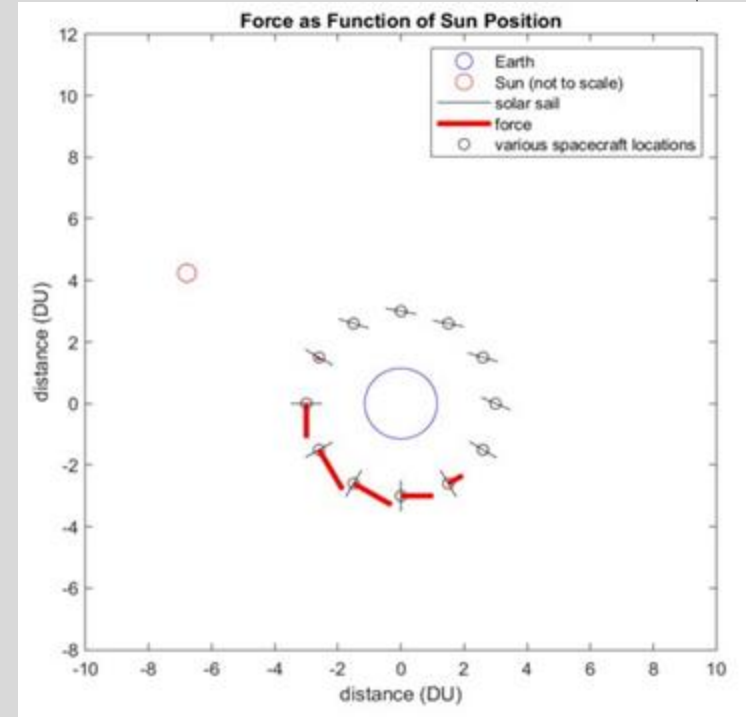


Figure 3: Visual representation of the force vector normal to the solar sail due to solar radiation pressure.

FORCING FUNCTION



Force is limited by the orbital energy of the spacecraft

- When spacecraft is close to Earth, Jacobi constant is limited to $C = -1.57$
- When spacecraft is close to the moon, Jacobi constant is limited to $C = -1.58609$

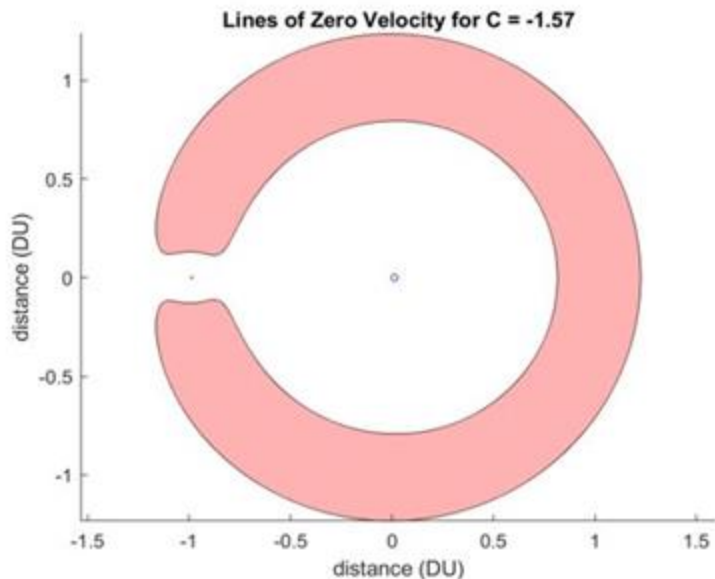
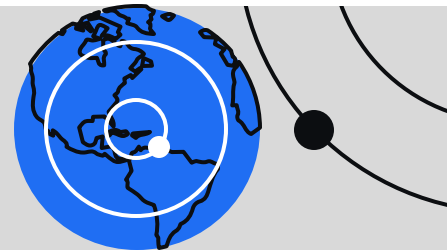


Figure 4:

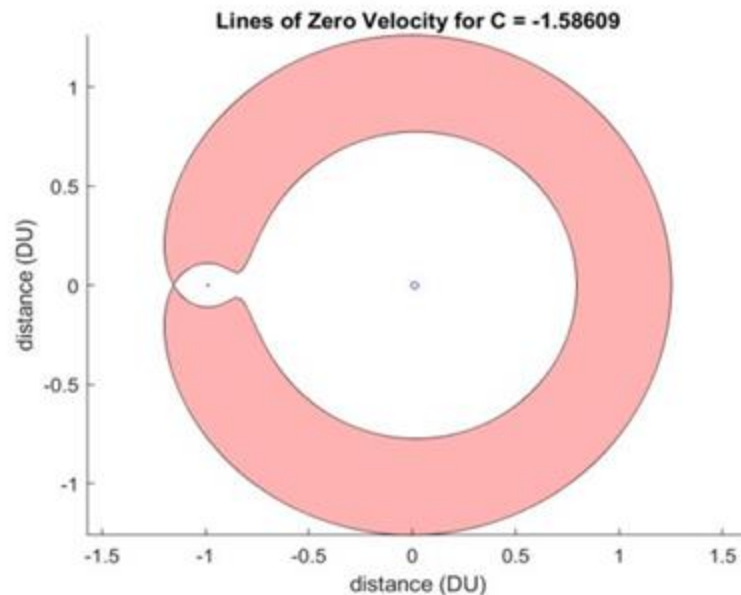
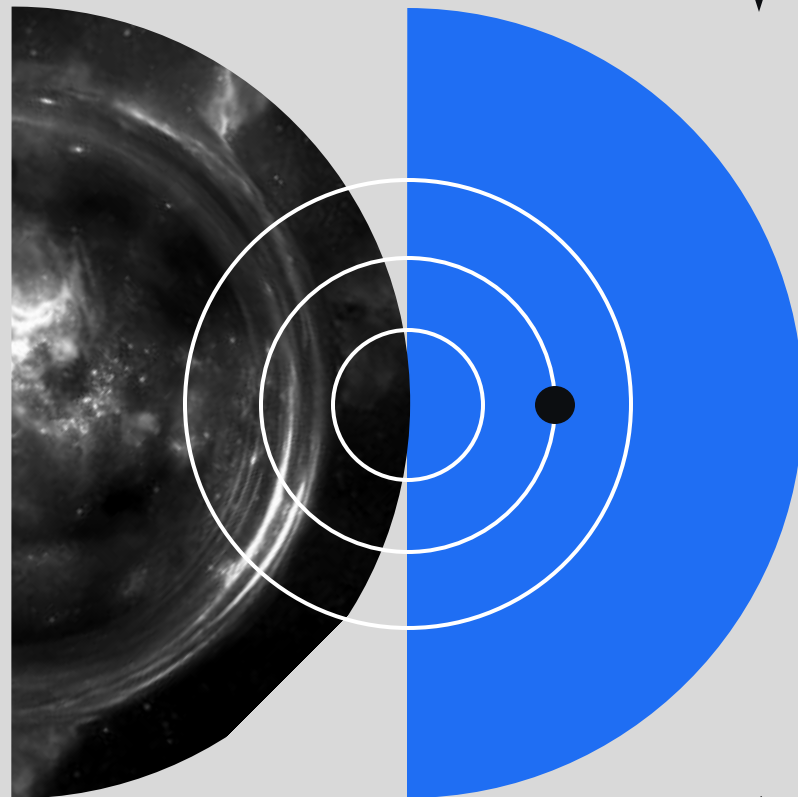


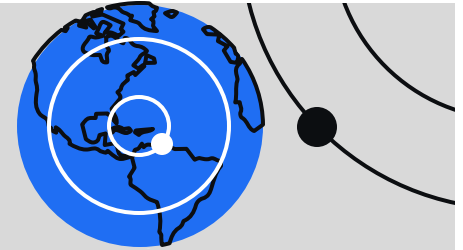
Figure 5:

03

RESULTS



RESULTS



Jacobi constant can be plotted over time for each simulation, providing an indication on when it reaches the critical value needed to reach the moon (-1.586)

- Results show that under a time constraint of 1 year, a solar sail area of **580 m²** is the minimum solar sail area required
- Convergence to required energy for $A = 1000 \text{ m}^2$ occurs at approx. 57 TU while $A = 600 \text{ m}^2$ occurs at approx. 81 TU

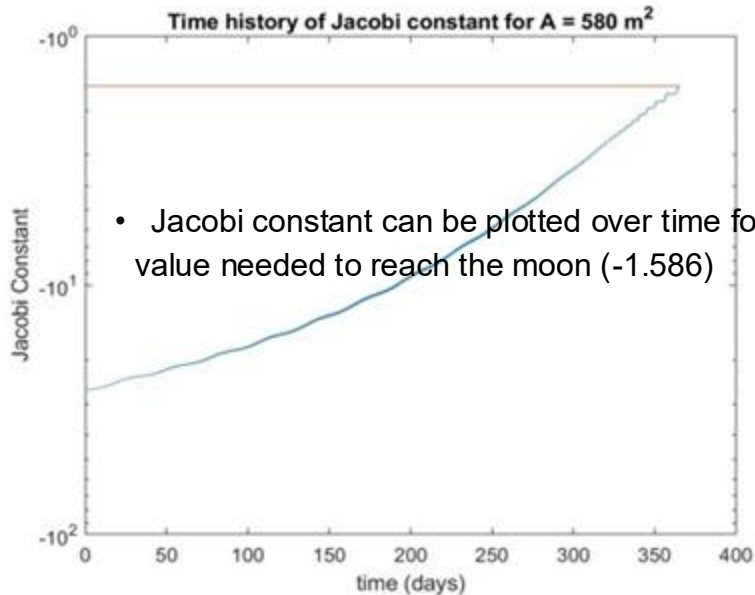


Figure 6:

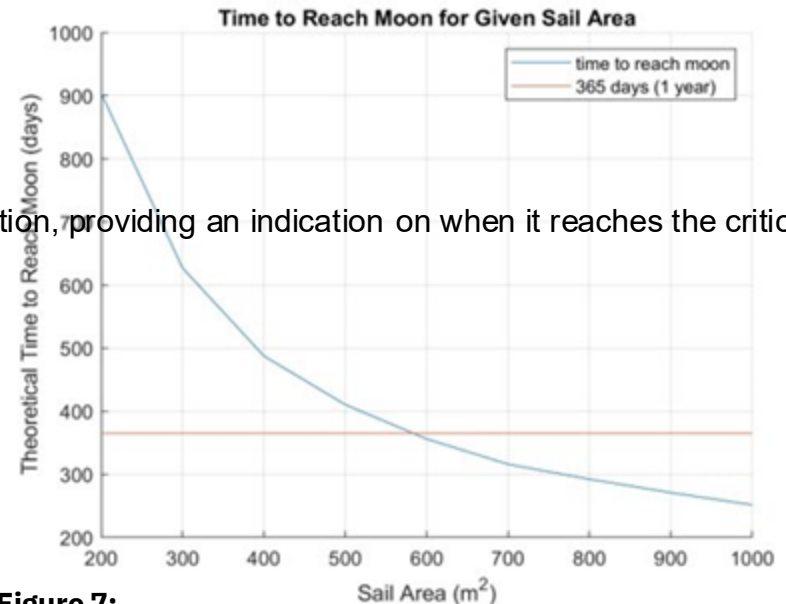


Figure 7:

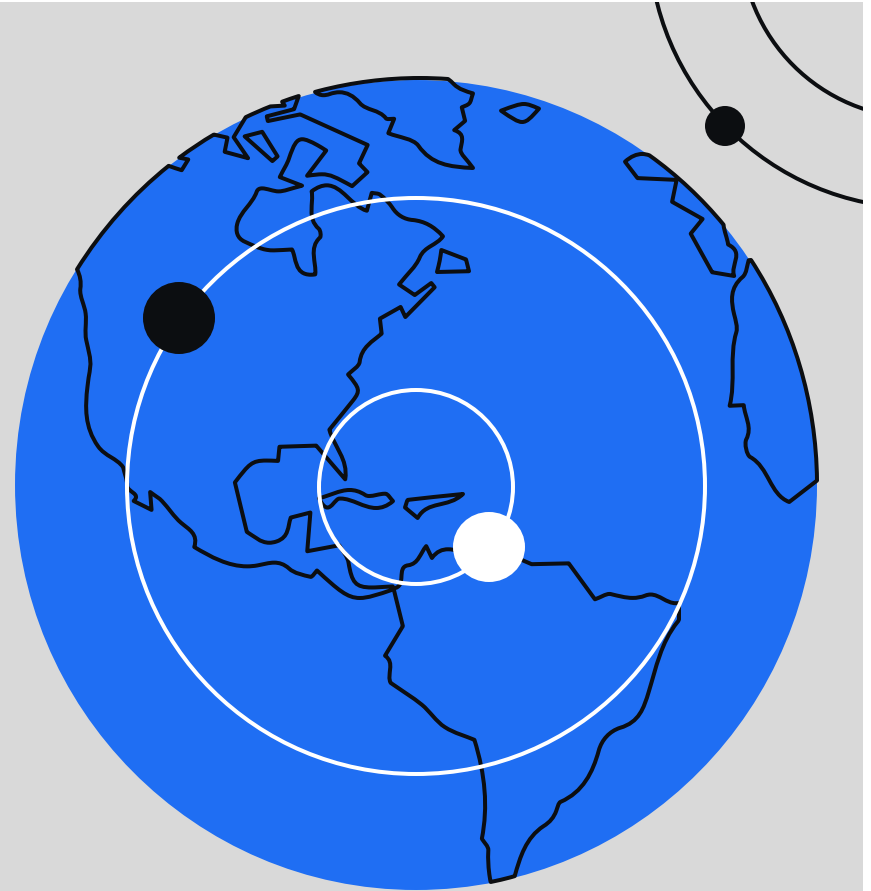
RESULTS



SPECIFICATIONS

$$V_{min} = t_{ss} \times A_{min}$$

- Solar sail thickness $t_{ss} = 2\mu\text{m}$
- Theoretical minimum volume of sail = 0.00116 m^3
- 1U of space typically allocated for electronics (0.001 m^3)
- Total impulse from earth to moon = 1143.8 Ns
- Thrust: 0 N (min in shadow) to $7.769 \times 10^{-3} \text{ N}$ (max) ✨

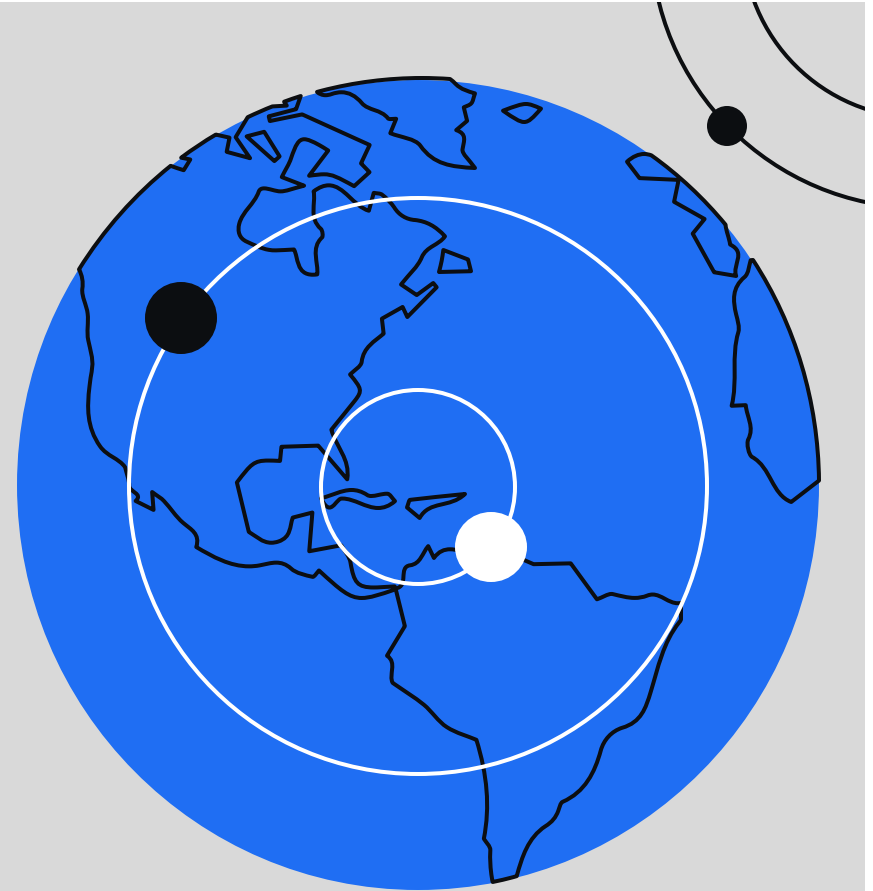


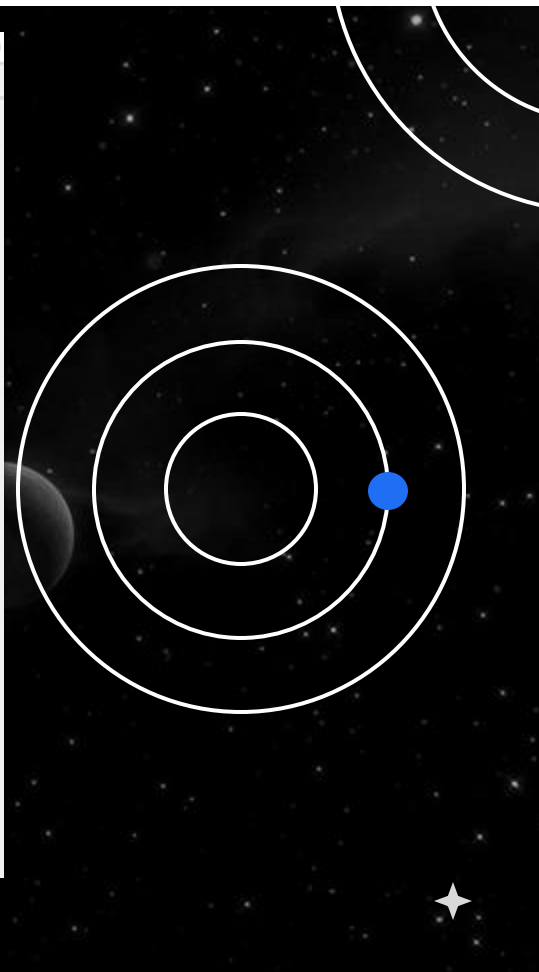
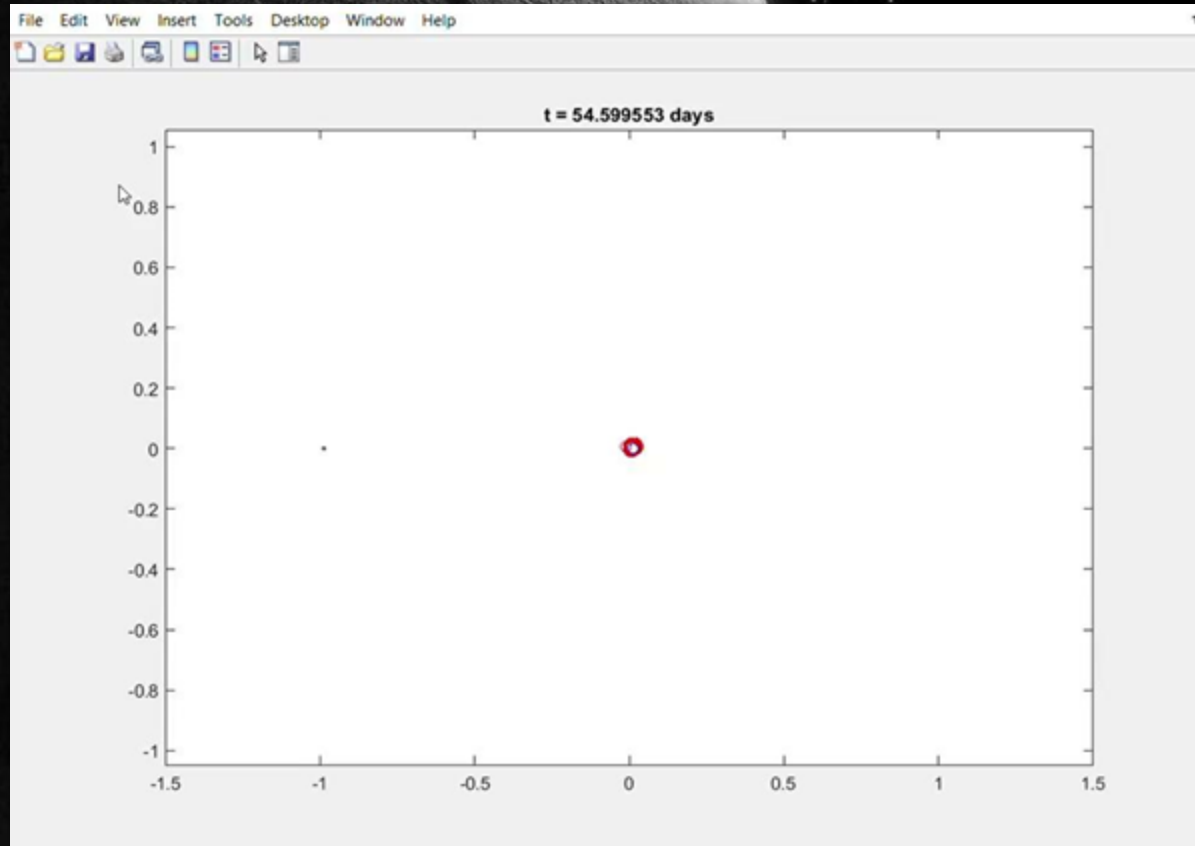
RESULTS



FORM FACTOR

- Based on these results, it can be concluded that **a 2U form factor is not possible**
- 4U is estimated to be the minimum CubeSat size
 - Account for smaller internal volume of CubeSat
 - Account for volume taken up by sail boom structures and deployment mechanisms
 - Account for packing efficiency of sail





RESULTS



- An example trajectory of an interplanetary solar sail might look as follows with a sail area of 1000 m² and mass of 5 kg reaching the moon in approximately 252 days
- Although the minimum area is 580 m², practicality of using this area is limited because it is difficult to align trajectory with gap in accessible regions. A larger sail area gives more time for trajectory to align with gap

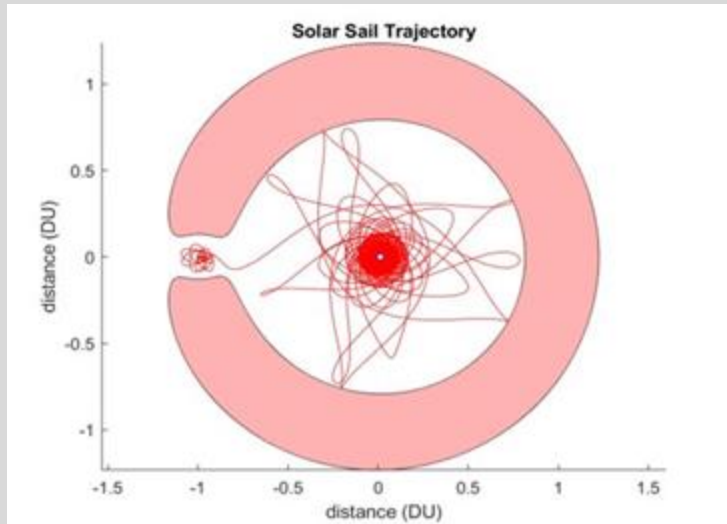


Figure 7:

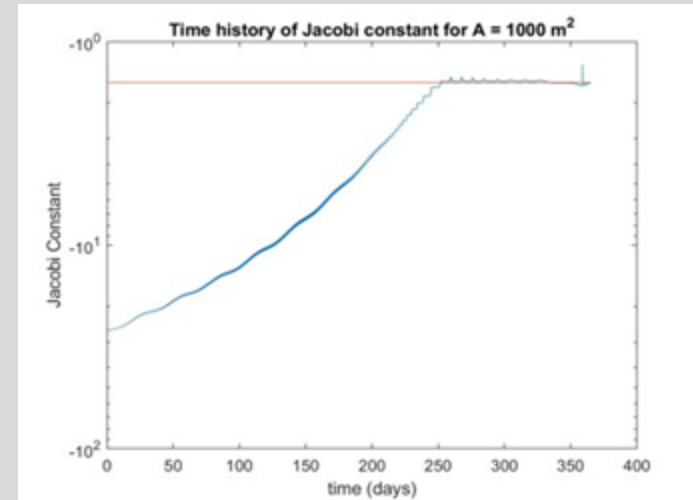
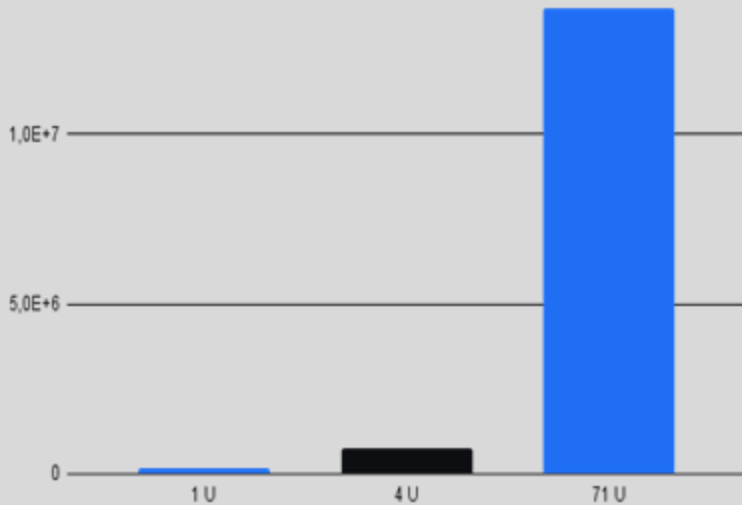


Figure 8:

NASA most recently sent a mission called CAPSTONE on a RocketLab rocket called electron on June 28, 2022. This is an excellent comparison. Our model is simpler compared to CAPSTONE (71 U vs 4 U), and is a steppingstone towards smaller CubeSat missions to other planets.



Actual cost/ predicted cost to make it interplanetary (approx. 2.03 times the typical)

1 U

Dimension

10 cm * 10 cm
* 20 cm

Typical cost

\$ 90,000

4 U

4*(10 cm *
10 cm * 20
cm)

\$ 360,000

**71 U
(NASA
Capstone)**

71*(10 cm * 10 cm
* 20 cm)

\$ 6,390,000



FUTURE WORK



01 FUTURE USE

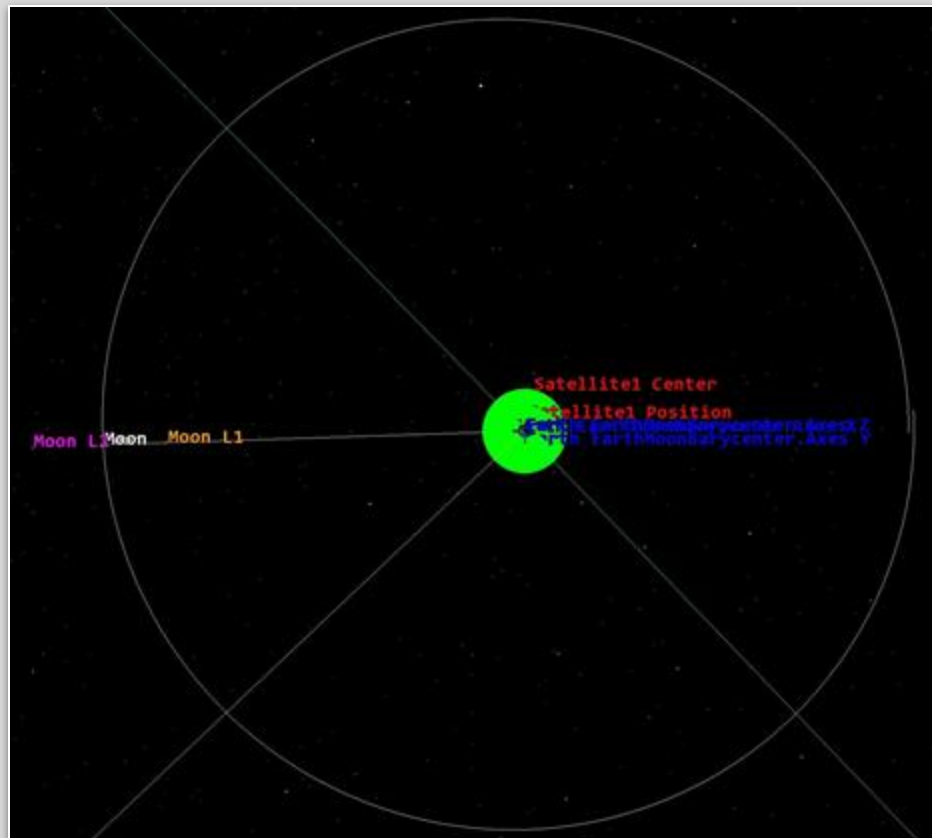
- The model built in MATLAB can be the basis for future technology demonstration missions of novel solar sail deployment mechanisms.



02 PROMISING STK SIMULATION

- An STK scenario was created using the assumptions and methods mentioned before, where we get a similar flower-like pattern around planet Earth, which validates that we are moving in the right direction. Further validation was done by modifying the MATLAB forcing function to get a similar trajectory. Although more testing can be done, initial testing shows that implementing a forcing model may be beyond the limits of STK software or that we don't understand stk's settings completely.





★ Figure 9:

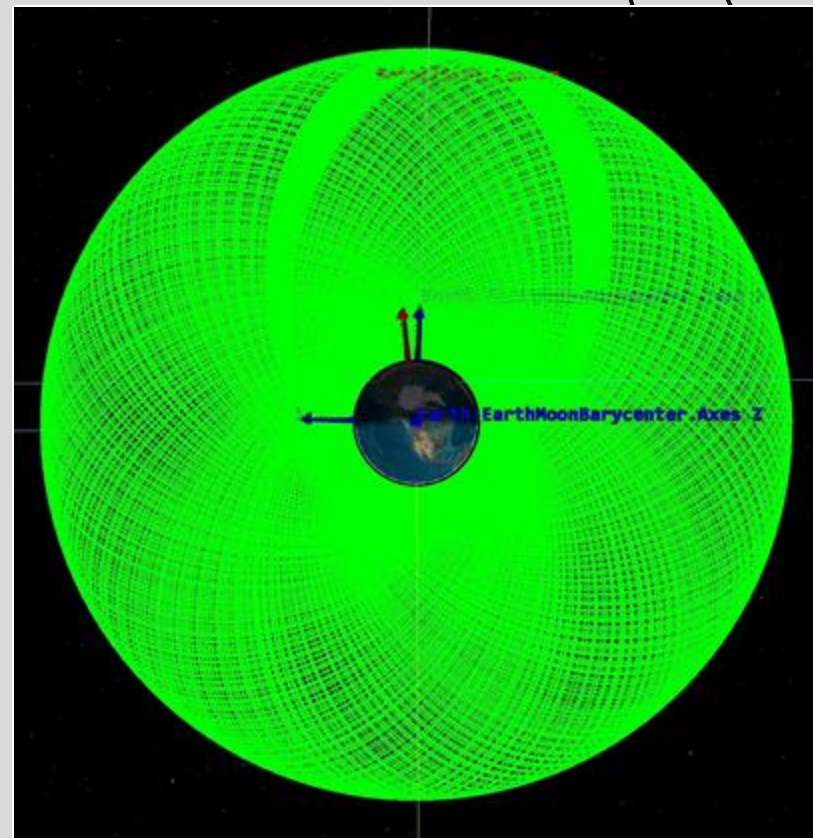


Figure 10:

03 **PHYSIOLOGY**

- Material and ability to get more thrust with a thinner solar sail.
- A thinner solar sail means a lot less strain energy while packing the sails in a compact form, hence a larger area can be obtained once expanded in space/orbit.

05 **3 BODY PROBLEM**

- Because of chaotic nature of 3 body problem, course corrections via PID control would need to be added for any practical implementation.

04 **STK**

- Adjustments to the STK scenario we might be able to achieve a similar trajectory. Although it is technically able to reach the moon, the trajectory just happens to not line up perfectly. For the Matlab model, Iterations over 360 degrees of initial spacecraft angles around Earth to find the initial conditions were executed that finally allowed the trajectory to line up with the gap and make it to the moon, a similar iterative process may have to be carried out with STK before we reach a similar outcome. There may be a faster way to do this with a better understanding of STK itself.

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